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Blockchain-Driven Decentralization for Data Sharing in the Metaverse

1st Saeed Hamood Alsamhi
Insight Centre for Data Analytics
University of Galway
Galway, Ireland
saeed.alsamhi@insight-centre.org

2nd Ammar Hawbani
School of Computer Science
Shenyang Aerospace University
Shenyang, China
anmande@ustc.edu.cn

3rd Abdu Saif
Faculty of Engineering and IT
Taiz University
Taiz , Yemen
sabdu.saif@taiz.edu.ye

4th Edward Curry
Insight Centre for Data Analytics
University of galway
Galway, Ireland
edward.curry@insight-centre.org

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***Abstract*—This paper presents a novel decentralized architecture that aims to transform the metaverse by smoothly incorporating blockchain technology. The proposed framework promotes decentralization principles to address centralization issues and restrict user empowerment in existing metaverse systems. It prioritizes putting people at the centre of their digital experiences. The decentralized framework goes beyond traditional tokenization methods by adding extra layers, like non-fungible tokens (NFTs) that change who owns an asset, and decentralized autonomous organizations (DAOs) that give people in communities more power, thereby fostering creativity and inclusion in virtual environments. Built upon the fundamental concepts of decentralization and user-centric design, the framework tackles the existing limits of the metaverse and envisions a future in which people actively define their virtual destiny. In addition to addressing issues, the framework provides possibilities for cooperative advancement, decentralized exchange of information, incorporation of future technology, and market growth, acting as a catalyst for innovation within the developing metaverse ecosystem. The framework embodies not only a framework but also a vision for a decentralized and user-centric metaverse, fostering creativity, inclusion, and the realization of virtual aspirations.**

***Index Terms*—Metaverse, Virtual Reality, Blockchain, Decentralized data sharing.**

I. Introduction

The emergence of the metaverse, a digitally immersive and interactive virtual domain, signifies a fundamental change in how people interact with digital material and socialize in virtual settings [1]. The metaverse has significant issues related to centralization, data ownership, and interoperability, which hinder its capacity to achieve real decentralization and empower users [2]. The present state of the metaverse is defined by the dominant control of platform operators over data ownership, management of virtual assets, and overall user experiences [4]–[6]. The concentration of power in one entity gives rise to issues over user independence and confidentiality and hinders the genuine democratization of the online realm. The drawbacks of centralized systems become evident as users navigate vast virtual worlds, emphasizing the need for a fundamental reconsideration of the metaverse framework [3], [7], [8].

Decentralized arises as a forward-thinking framework based on decentralization, transparency, and user-centric design to address these difficulties. The principle of decentralization is the governing philosophy, in line with the concept of a metaverse where people actively choose their digital futures [9]–[12]. The framework employs blockchain technology to actualize this goal, offering a secure, transparent, and customizable foundation for transforming digital interactions in the metaverse [13]. The selection of blockchain technology is not random; instead, it is a deliberate reaction to the metaverse's fundamental need for systems devoid of trust that can be verified. Blockchain's decentralized ledger keeps transparent, unchangeable records open to everyone, changing virtual asset ownership and transforming metaverse transactional procedures [14].

The decentralized develops metaverse applications and experiences by addressing current problems and implementing interoperability standards that enable the smooth exchange of assets across different metaverse ecosystems. The non-fungible tokens (NFTs) aim to transform the ownership of digital assets [15] and promote user ownership and control of the virtual environment by incorporating decentralized autonomous organizations (DAOs), which improve the capabilities of metaverse communities. Integrating DAOs in design is used to increase people's power in the metaverse, in which users engage in the governance and decision-making processes. Users have to affect how the virtual environment is managed, how updates and changes are implemented and resources allocated. Zero-knowledge proofs (ZKP) enhance security and privacy in the metaverse by guaranteeing private information is kept secret. It helps resolve privacy concerns and guarantees that users engage within a highly secure virtual environment. Privacy and security in virtual transactions is the rationale for integrating

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zero-knowledge proofs into transactions and creating a reliable and secure metaverse environment [24].

By examining previous studies on digital currencies, blockchain and artificial intelligence (AI) in the virtual world, and metaverse components, the authors of [16] investigated the metaverse and the relationship between blockchain and AI. Additionally, integrating blockchain technology and intelligent networking—including communication and computation networks—into the metaverse and the potential and challenges associated with this approach are covered in [17]. Some common metaverse applications are examined to bridge the gap between academic research findings and practical products [18]–[20]. Recognizing the importance of blockchain technology for the metaverse, the authors of [21] presented a comprehensive overview that elucidates the function of blockchain technology in the metaverse, which includes a detailed examination of digital asset management. Given that Industry 5.0 is a subset of Society 5.0, the implications of blockchain technology for resolving the increasing cybersecurity challenges toward safe and intelligent production are described [22], [23].

The strategic advancement of the decentralized framework arises from a profound understanding of the deficiencies in the the present metaverse structures. The concept aims to redefine the metaverse as a decentralized and user-centric environment, focusing on resolving data management and user autonomy problems. This realization stems from acknowledging that complete decentralization has yet to be achieved [25]. The incorporating blockchain into the metaverse is likewise addressed by the architecture including privacy, governance, and cross-system communication capabilities [25], [26]. The framework is designed to address the present limitations in user engagement by promoting active user participation to realize decentralized aspirations and create a user-driven, inventive metaverse [1], [26]. Our framework offers a unique perspective on user-centric design and blockchain integration while introducing novel elements,i.e., NFTs and DAOs, to empower users and address existing data ownership and management limits. The contributions of this paper are summarized as follows:

1) **Holistic Blockchain Integration:** The novel framework integrates blockchain with the metaverse to address privacy, governance, and interoperability. The framework emphasizes user-centricity and strengthens the metaverse to enable people to interact, guarantees safe transactions with privacy protections, and makes asset trading across several virtual worlds.
2) **Empowering User-Driven Experiences:**The decentralized frameworks empower users in the metaverse by including DAOs; the design encourages community involvement and allows users to actively engage in decision-making and shape the virtual space. The framework uses NFTs and emphasizes user security and privacy to reinvent the idea of digital asset ownership. Therefore, the framework guarantees that users will mold the metaverse to suit their preferences. By emphasizing user-driven experiences, the framework becomes a potent catalyst for developing decentralized ideals inside the metaverse, marking a dramatic shift in how virtual worlds are seen.

II. Decentralized Metaverse Framework

The decentralized framework architecture incorporates blockchain into the construction of the metaverse ecosystem, including metaverse, blockchain, asset ownership and interoperability, user empowerment, privacy and security, and real-world data integration layers (as shown in Fig.1). The digital worlds, interactive elements, and user experiences that comprise the metaverse canvas are part of the metaverse layer [29], [30]. Building on this idea, the blockchain layer uses a distributed ledger to offer transparency and security by connecting with the metaverse layer.

The privacy and security layer addresses challenges in the metaverse by protecting user identities using zero-knowledge proofs [31]. Integrating external data utilizing blockchain oracles, the real-world data integration layer improves virtual experiences and virtual environments. In decentralized environments, users control how their virtual experiences [14]. Integrating blockchain and metaverse, the framework opens up new possibilities and ushers in a new era of community involvement, user empowerment, and innovation inside virtual environments.

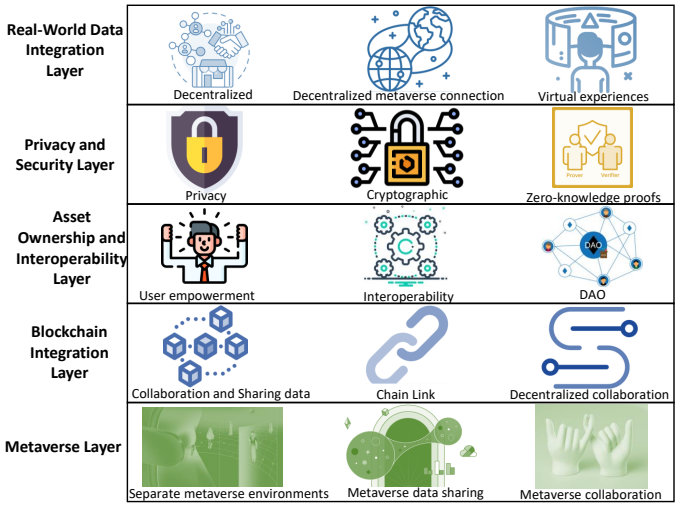

Fig. 1. Proposed Framework for enabling data sharing of Metaverses

A. Metaverse Layer

The metaverse layer is a framework component determines the nature of the virtual world and user interactions in the metaverse. The layer in question extends beyond a single point of emphasis, including digital realms, virtual possessions, and interactive components, all of which enhance the immersive quality of the metaverse. The layer in question is represented as a canvas and plays a crucial role in determining the architecture and user interactions. The metaverse layer is the core element of user involvement and immersion, influence on users' perception and interaction in the metaverse.

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B. Blockchain Integration Layer

The blockchain integration layer plays a crucial function by directly connecting with the metaverse layer, serving as the middleman that smoothly incorporates distributed ledger technology into the metaverse architecture. The selection and implementation of an appropriate blockchain platform, such as Ethereum or Binance Smart Chain, is essential for supporting the features of the metaverse. When choosing, evaluating scalability, consensus processes, and ecosystem support is crucial since the variables form the basis of the metaverse. Deploying smart contracts is a crucial element of the blockchain integration layer to manage interactions inside the metaverse. Smart contracts are contractual agreements. The blockchain layer facilitates decentralized governance by deploying DAOs to enable users to actively influence the virtual environment inside the metaverse. Integrating the blockchain represents change, in which people have a regulating and contributing to the metaverse environment.

Decentralized data sharing is facilitated by enabling technologies such as blockchain, which guarantees the immutability and openness of data. Furthermore, smart contracts are used for automated and trustless transactions, as seen in Fig.1. The technologies allow users to safely and openly exchange data without relying on a central authority, promoting innovation in many applications, such as blockchain-based platforms. The figure shown in Fig.2 illustrates a situation where four separate metaverse instances run on their server. It also presents the blockchain as the connecting layer that promotes decentralized cooperation. Each metaverse prioritizes decentralization by operating independently on dedicated servers. The blockchain layer is a technical link that facilitates communication, asset exchange, and interactions between various virtual worlds. The integration is expected to improve efficiency and effectiveness by simplifying operations using smart contracts and maintaining consistent ownership of assets across different metaverses. The decentralized structure adheres to the fundamental principles of blockchain and creates a resilient and interconnected metaverse ecology. Nevertheless, it is essential to consider scalability and interoperability standards to ensure optimal functionality of this collaborative framework.

Intgrating th blockchain layr into a currnt mtavrs implmntation involvs svral ky stps for samlss intgration. Firstly, it is crucial to evaluate the compatibility of the existing maintenance infrastructure with blockchain technology, considering factors like architecture and functionality. Then, a suitable blockchain platform such as Ethereum or Binance Smart Chain based on scalability, consensus consnsus mchanisms and smart contract capabilities is essential. Dsigning tailored smart contracts to facilitate various mtavrs functionalities like assistant ownership and government management mechanisms follows suit. Integrating blockchain nodes into the mtavrs infrastructure enables communication with the chosen blockchain platform and supports data storage, transaction processes and consensus mechanisms. It configures appropriate consensus mechanisms, like Proof of Stak or Dlgatd Proof of Stak, and ensures transaction security and integrity. Providing intuitive user interfaces and tools for smooth user engagement with blockchain features in the metaverse environment is crucial. Finding problems is aided by extensive training in a controlled setting into the living world, which adding new features, improved security, transparency, and user empowerment within the virtual environment, ongoing monitoring, maintenance, and upgrades after deployment, guarantee optimal performance.

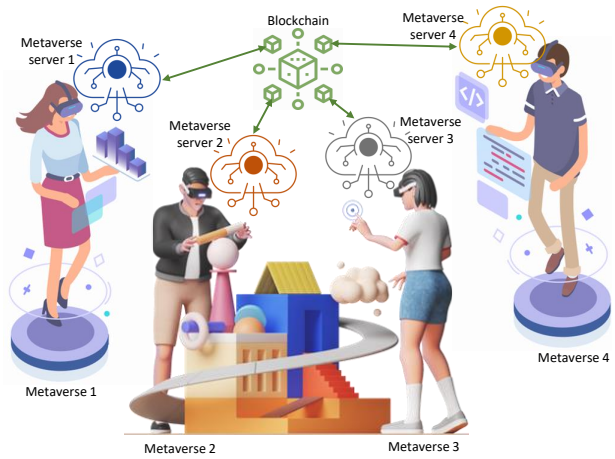

Fig. 2. Blockchain enabled decentralized metaverse [27]

C. Asset Ownership and Interoperability Layer

The Blockchain Integration Layer is the foundation for the asset ownership and interoperability layer to change the metaverse environment. Asset ownership and interoperability layer's seamless integration of NFTs transforms ownership of digital assets. In the metaverse, NFTs are cryptographic tokens that let users prove ownership and authenticity. By including interoperability standards beyond individual ownership, the layer broadens the scope of the metaverse experience. The standards provide an interconnected virtual ecosystem by facilitating the seamless movement of digital assets between metaverse settings. Thanks to NFTs and interoperability, users travel and communicate with many metaverse platforms, making the virtual world active and profitable. A step toward a metaverse that puts user requirements first and encourages cooperation is the inclusion of NFTs. NFTs offer advantages in the context of the metaverse, such as enabling users to safely purchase and sell digital assets within a virtual environment; enhancing the value and legitimacy of virtual assets by ensuring scarcity; and enabling interoperability standards that make for digital assets to flow between different metaverse ecosystems.

D. Privacy and Security Layer

To maintain the integrity of the whole system, the privacy and security layer plays a crucial role. Its primary purpose is to develop mechanisms that protect user privacy and strengthen the security of transactions in the metaverse. This layer is crucial in ensuring security using advanced techniques, mainly

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focusing on implementing zero-knowledge proofs. ZKP secures transactional data and user identities in the metaverse setting. ZKP validates interactions and transactions in the metaverse without disclosing private information about users of the transactions. The layer utilizes zero-knowledge proofs and cryptographic algorithms, allowing safe validation without disclosing sensitive information. The implementation is to protect user identities and transactional data from attacks, guaranteeing high secrecy to establish a safe metaverse environment.

E. Real-World Data Integration layer

The real-world data integration layer is the decentralized framework's architecture, which utilizes blockchain oracles to seamlessly transfer real-time data from the real world to the metaverse. This layer improves virtual experiences by linking the decentralized metaverse environment and the real world. Using real-world data enhances the virtual environment with contextual information. The framework's design is organized, starting with the core metaverse layer and advancing via blockchain integration, asset ownership, user empowerment, privacy and security, and finally reaching the real-world data integration layer. The advancement of the strata demonstrates metaverse advancement that tackles crucial concerns, including user engagement, asset ownership, security, and the seamless integration of real-world components.

Summary: Th cor of th framework lis th mtavrs layr and srving usr intractions and immrsiv virtual xprincs. The blockchain integration layer connects with the medium layer and incorporates a distributed ledger to provide transparency, security, and programmability. Th asst ownrship an' introprability layr revolutionizes digital asst ownrship by simply incorporating NFTs and introprability standards and facilitating smooth asst transfr across different mtavrs platforms. The user layr enables active user participation in decision-making processes. The privacy and security layr employs advanced technologies (i.e., zero-knowledge proofs to protect user identities and transactional data and ensure a safe and secure mtavrs environment). Lastly, the real world data integration layr utilizs blockchain oracls to simply transfer real world data to the mtavrs, enhancing virtual experiences with contextual information and bridging the gap between the dcntralizd mtavrs environment and external data sources. This comprhnsiv dsign strategy aims to rinvnt th mtavrs by samlssly intgratin' blockchain technology and addressing crucial concerns, advancing user engagement and asst ownership and security, and integrating real-world components.

III. Discussion and Validation

A. Evaluating the Framework's Effectiveness

The decentralized architecture offers an innovative and all-encompassing method for incorporating blockchain technology into the metaverse. When assessing its efficacy, numerous crucial factors become prominent. Incorporating blockchain technology effectively provides transparency, security, and programmability inside the metaverse, per the ideals of decentralization [14]. Incorporating NFTs and interoperability standards into the asset ownership and interoperability layer significantly transforms the landscape of digital asset ownership. This advancement enables users to transfer assets across various metaverse ecosystems effortlessly. The User Empowerment Layer enables users to participate in decision-making by installing DAOs and promoting community ownership. Table.I Illustrated the comparison of decentralized versus centralized metaverse.

Privacy, security, and real-world integration: Incorporating zero-knowledge proofs into the privacy and security layer is essential for effectively resolving problems related to data management and security problems in the metaverse. This layer enhances the integrity and safety of the virtual world, guaranteeing that users may participate confidently. The Real-World Data Integration layer enhances virtual experiences by connecting the decentralized environment and real-world data via oracles. This integration brings a level of realism and relevancy, which improves the entire user experience.

Addressing existing gaps: The decentralized framework tackles the deficiencies in current metaverse systems, offering answers to centralization, asset ownership, and user issues. The paradigm intrdouces metaverse experience because of the changing expectations of a digitally empowered user. Future framework versions include incorporating developing blockchain in privacy-preserving techniques by exploring additional layers or modifications.

Implications for Metaverse: The decentralized framework establishes a safe, efficient, inclusive, and dynamic metaverse by promoting decentralization, transparency, and user empowerment. It establishes a standard for advancing decentralized metaverse, shaping the virtual environments to promote user independence, ingenuity, and cooperation.

Limitations and considerations: The decentralized structure promises to understand the ability to manage the growing needs of integrating blockchain and the requirement that the broader metaverse community accept decentralized concepts.

B. Validation

The selected blockchain platform was set up to accommodate the suggested design, with smart contracts implemented to regulate interactions and transactions. The framework's performance in crucial domains such as asset ownership, user empowerment, privacy, and security was evaluated by simulating realistic situations.

Asset Ownership and Interoperability: The validation procedure primarily focused on the asset ownership and interoperability layer, scrutinizing how the framework redefines digital asset ownership via NFTs and enables the seamless exchange of assets across different virtual worlds. User involvement in asset transfer procedures was monitored, and an assessment was made of the efficacy of interoperability standards in facilitating smooth exchanges across various virtual environments.

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TABLE I
COMPARISON OF DECENTRALIZED VERSUS CENTRALIZED METAVERSE

Feature	Decentralized	Centralized
Control and Governance	Distributed among users	Controlled by a single entity
Security and Privacy	Efficient	Depending on the centralized entity
Scalability	Achieved through distributed and P2P networks	Limited by the centralized server capacity
Innovation	Foster more innovation	Limited by entity's priorities and restrictions
Autonomy	Limited autonomy for users; dependent on central entity's policies and rules	Users have greater autonomy and agency in shaping the virtual world and participate in community-driven governance
Transparency	Emphasis on transparent data management and accountability through decentralized protocols	Lack of transparency in decision-making and data management
Regulatory challenges	Compliance with regulations, potential for monopolistic practices	Navigating regulatory landscape of blockchain and distributed technologies, addressing legal and policy considerations

User Empowerment and Governance: The actual implementation of DAOs verified the user empowerment layer. Metaverse users were given chances to participate in decision-making procedures, and the influence of DAOs on community involvement and governance frameworks was noted. Feedback mechanisms and user satisfaction surveys were used to assess the perceived level of empowerment among users.

Privacy and Security Measures: The privacy and security layer evaluate via the surveillance of transactions and user engagements in the metaverse. The zero-knowledge proofs in safeguarding user identities and transactional data have been verified. Security and penetration testing were assessed to detect and rectify any weaknesses.

Real-World Data Integration: The performance of the real-world data integration layer was assessed by integrating blockchain oracles to retrieve external information . An evaluation was conducted to determine the precision and promptness of incorporating in a metaverse environment. The assessment is considered the dependability of the data, the transmission delay, and user experiences.

Results and Findings: The validation method yields findings across aspects. Users exhibited an increased perception of ownership and control over digital assets, indicating the NFT with implementing DAOs to enhance user experience. Privacy and security safeguards, such as zero-knowledge proofs, effectively safeguarded user data and transactions. Integrating real-world data has shown the ability to enhance virtual experiences by providing up-to-date and pertinent information.

Limitations and Future Directions: While there have been excellent results, it is crucial to recognize the constraints of the validation procedure. Issues about scalability, particularly in expansive metaverse settings, were recognized and provided opportunities for future enhancement. Continual user input and iterative enhancements will be essential in perfecting the framework for broader acceptance and tackling emergent obstacles in the changing metaverse environment.

IV. CHALLENGES AND OPPORTUNITIES

A. Challenges

Scalability: Although the decentralized architecture has shown encouraging validation findings, its scalability remains a crucial obstacle to its more comprehensive implementation. The expansion of the metaverse creates worries regarding transaction throughput and network congestion as it accommodates a growing user base and many decentralized apps (DApps) within the intended architecture. In order to tackle these issues, it is imperative to investigate scaling solutions, such as layer 2 protocols.

User Adoption and Education: The system's effectiveness is highly dependent on the extent to which users accept and comprehend decentralized concepts. Mainstream user adoption may need help mastering the intricacies of blockchain technology and the metaverse. In order to close this gap, it will be crucial to implement strategic educational activities and develop user-friendly interfaces.

Interoperability Standards: The framework presents interoperability standards, but the issue lies in attaining extensive industry acceptance of these standards across many metaverse platforms. Engaging in cooperative initiatives with industry players and developing generally recognised standards will be essential to enable the smooth transfer of assets across various virtual environments.

B. Opportunities

Collaborative Development and Standards Setting: The constraints of achieving interoperability and scalability in the blockchain and metaverse domains provide prospects for collaborative growth within the respective communities. Participating in open-source initiatives and helping establish industry standards may promote a collaborative endeavour to overcome common obstacles. **Emerging Blockchain Technologies:** The fast development of blockchain technology presents prospects for incorporating nascent solutions into the architecture. By investigating innovations like sharding, sidechains, and consensus methods, it is possible to improve scalability and performance, thereby assuring that the framework can quickly react to environmental changes.

Community Engagement and Governance: The User Empowerment Layer facilitates ongoing community interaction and enhances governance processes. The potential exists in enhancing DAO methods, integrating user feedback loops, and actively enabling the community to influence the development of the metaverse. This interactive method enhances user involvement and adds to the improvement of the framework. **Market Expansion and Adoption Incentives:** With the increasing popularity of the metaverse, new prospects for ex-

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panding the market emerge. Strategic alliances, incentivization initiatives, and user engagement campaigns may enhance the acceptability of the framework inside the metaverse ecosystem. Establishing the framework as a standard in decentralized metaverse development will need a solid and active user base, as well as providing incentives for involvement.

V. Conclusion

In this paper, the framework has presented an innovative and forward-thinking system that aims to revolutionize the metaverse by smoothly incorporating blockchain technology. The framework promotes decentralization principles to address centralization issues and restrict user empowerment, prioritizing users' control over their digital experiences. The architecture establishes a transformational metaverse by including novel layers such as redefined asset ownership with NFTs and community participation with DAOs. The promise extends beyond reducing present restrictions, providing chances for collaborative development, integrating upcoming technologies, and expanding the market. As the metaverse progresses, this architecture acts as a catalyst, stimulating innovation and preparing for a decentralized, user-focused digital future where people actively create and contribute to their virtual surroundings.

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